

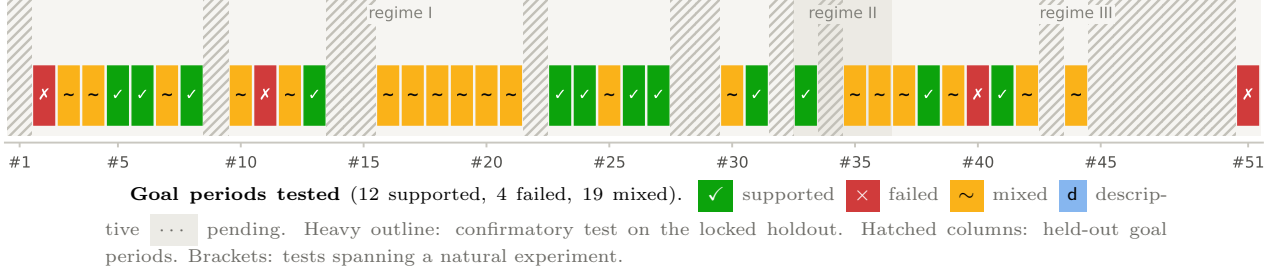
H19: One curve for all periods: per-period loop gains collapse onto a function of an operational control parameter

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Loop gains do not collapse onto one curve: activity co-activation and talk triggering are separate channels that moved in opposite directions across the scaffold switch.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

Pooled 244 per-period loop-gain estimates (9 methods from H02–H05 plus its own equal-time estimator) over 35 non-holdout periods, and asked whether they fall on one curve of x_{att} , attended partners per unit attention. **The pre-registered collapse failed:** equal-time activity gains rise with x_{att} (slope $+0.13$ [$0.06, 0.20$]) while Hawkes talk triggering falls (fast n_x : -0.06 [$-0.10, -0.02$]); out of sample a regime-only step beats the curve by 11.5 nats (era and $\log N$ also win). The opposite slopes trace the 2026-03-24 scaffold switch. **The unfitted mapping P3 holds:** the equal-time talk gain tracks the gain predicted from Hawkes fast kernels ($\rho = 0.44$, $p = 0.009$) and exceeds it in 91% of periods. Per-pair triggering scales as $1/(N - 1)$, not $1/(1 + \bar{k})$ (P4 failed). Post hoc, all 5 talk-channel gains rise with messages delivered per LLM step (+25 nats over regime-only); frozen for the holdout, not run.

PHYSICAL MODEL

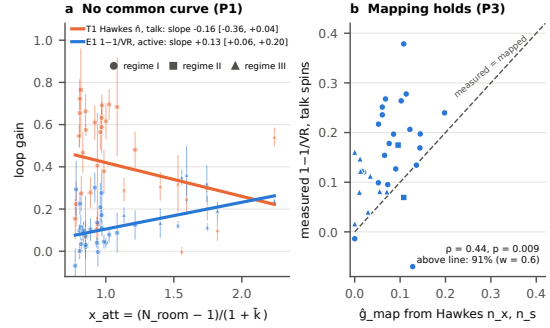
Models 01/02 (Ising) and 09 (Hawkes) in mean field. s_i : agent i 's 1-min active or talk spin; VR: variance ratio; g_{eq} : equal-time loop gain (H02's $\beta J_0 q$, H04's K , H05's g). n_x, n_s : Hawkes cross-/self-offspring per event; w : share within one $\Delta = 60$ s bin (kernel time τ_k). Dilution: per turn, $\bar{k}$ unread messages from $N_{room} - 1$ partners, response budget b . y_{pm} : method m 's gain in period p .

$$g_{eq} = 1 - 1/VR, \quad x_{att} = (N_{room} - 1)/(1 + \bar{k})$$

$$VR = \text{Var}(\sum_i s_i) / \sum_i \text{Var} s_i, \quad n_x = b x_{att}$$

$$\hat{g}_{map} = \frac{2wn_x}{1 + 2w(n_x + n_s)}, \quad w = 1 - \frac{\tau_k}{\Delta} (1 - e^{-\Delta/\tau_k})$$

$$y_{pm} = \alpha_m + c_m x_{att,p} + u_p + \varepsilon_{pm}, \quad \text{collapse: } c_m > 0 \forall m$$



Observables. (a) The two primary loop gains per period against x_{att} , with their random-effects meta-regression lines: Hawkes talk branching $\hat{n}$ (orange) falls while the equal-time active-spin gain (blue) rises, so no common curve exists; high x_{att} is essentially regime III (triangles). (b) The one unfitted cross-method prediction: the measured equal-time talk gain against the value mapped from H03's fast Hawkes kernels ($w = 0.75$). Points sit above the identity line: common fast fields add to triggering.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

The negative result is the useful part: a swarm has no single “coupling strength” to forecast from its configuration. Co-activation and reply triggering are different quantities that moved in opposite directions when the scaffold changed, so a dashboard reporting one “loop gain” would mislead. That is a warning, not a lever. P3 is reassurance that Hawkes and Ising estimates of talk coupling agree up to a known map. The post hoc curve (more chat exposure per model call, more reply triggering) is the one possibly actionable knob, but it was picked from 16 controls after the fact and is entangled with the scaffold; the rest is meta-analysis bookkeeping.

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